

DUL-99 — Sylhet Limestone Lost Circulation Knowledge Pack

Synthetic historical offset-well record for mud-loss retrieval and formation-specific context

IMPORTANT: This document is synthetic demonstration data prepared for the NWIS/SIH prototype. It must not be represented as an authentic confidential OIL record, real incident report, or verified SPE publication.

Well & Event Summary

Field	Value
Well	Duliajan-99 (DUL-99)
Field	Duliajan Field
Coordinates	26.848°N, 95.332°E
Loss depth	3210 m
Formation	Sylhet Karst Limestone
Hole size	8.5 inch
Bit	PDC 5-blade
Pre-loss mud weight	1.32 g/cc

Incident

At approximately 3210 m, the synthetic record describes total loss of returns in a naturally vuggy/fractured Sylhet Limestone interval. The annular fluid level dropped and the condition was classified as severe lost circulation.

Observed Indicators

Indicator	Synthetic observation
Returns	0 / 100% lost
Annular fluid level	Dropped ~120 m
Formation	Vuggy/fractured limestone
Mud weight before loss	1.32 g/cc

Mitigation

Step	Action
1	Top up annulus as required
2	Pump 60 bbl LCM pill
3	LCM blend: Nut Plug / Mica / CaCO3
4	Place 35 bbl Class-G cement plug across 3200–3210 m
5	Allow ~12 h set time
6	Confirm return restoration before proceeding
NPT	44 hours
Material cost	■18.5 lakh — synthetic demonstration value

RAG Metadata

Search terms: DUL-99, Sylhet, 3210 m, lost circulation, vuggy limestone, LCM, cement squeeze, zero returns. This record lets the assistant retrieve an analogous historical loss event when live flow-in/flow-out or pit-volume data indicate possible losses.

Synthetic Sensor Feature Example

timestamp	depth_m	flow_in_lpm	flow_out_lpm	pit_volume_bbl	formation
09:00	3206	1800	1790	520	Sylhet Limestone
09:05	3207	1800	1600	518	Sylhet Limestone
09:10	3208	1800	1200	514	Sylhet Limestone
09:15	3209	1800	600	505	Sylhet Limestone
09:20	3210	1800	0	490	Sylhet Limestone